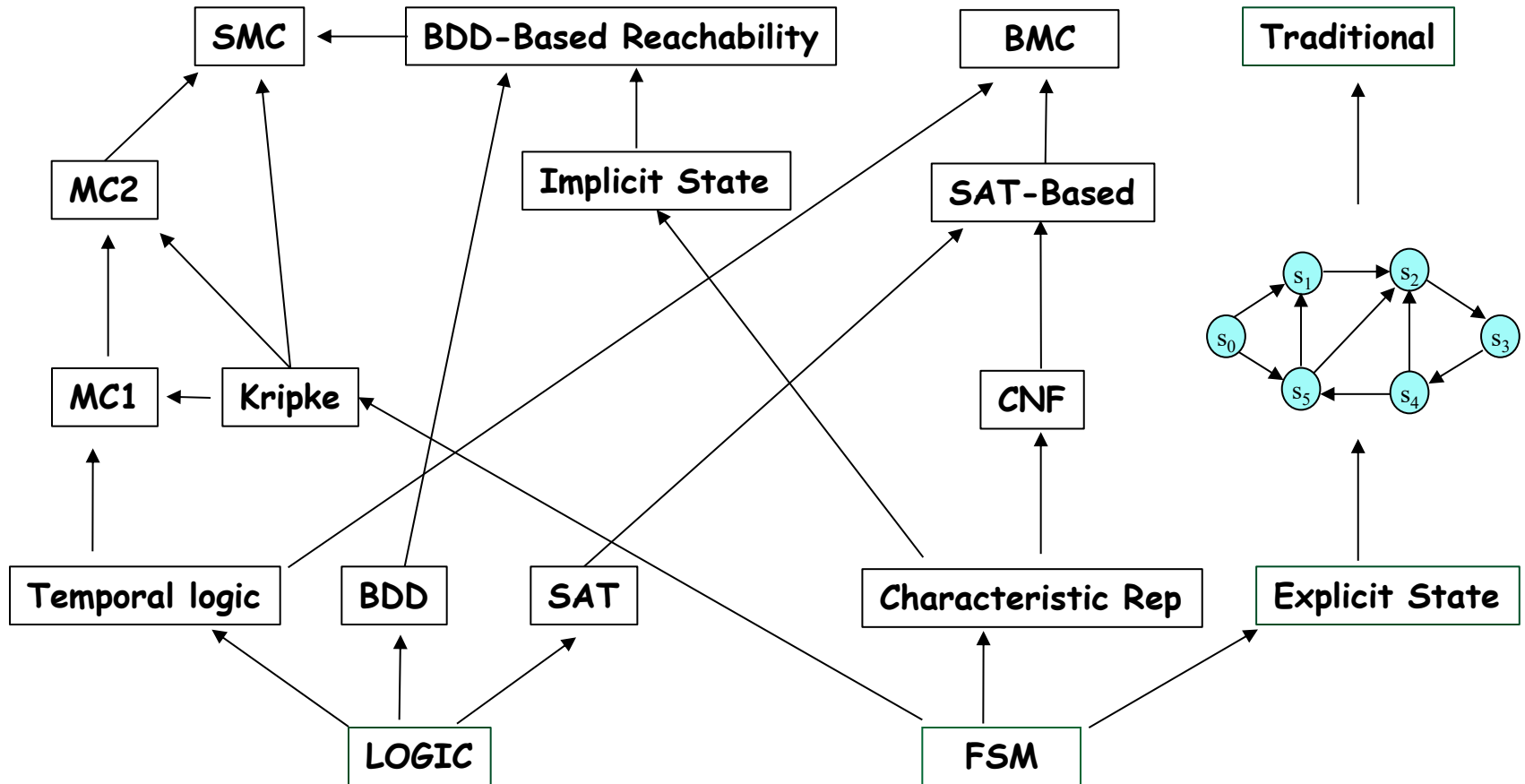




Characteristic Representations for Combinational Circuits

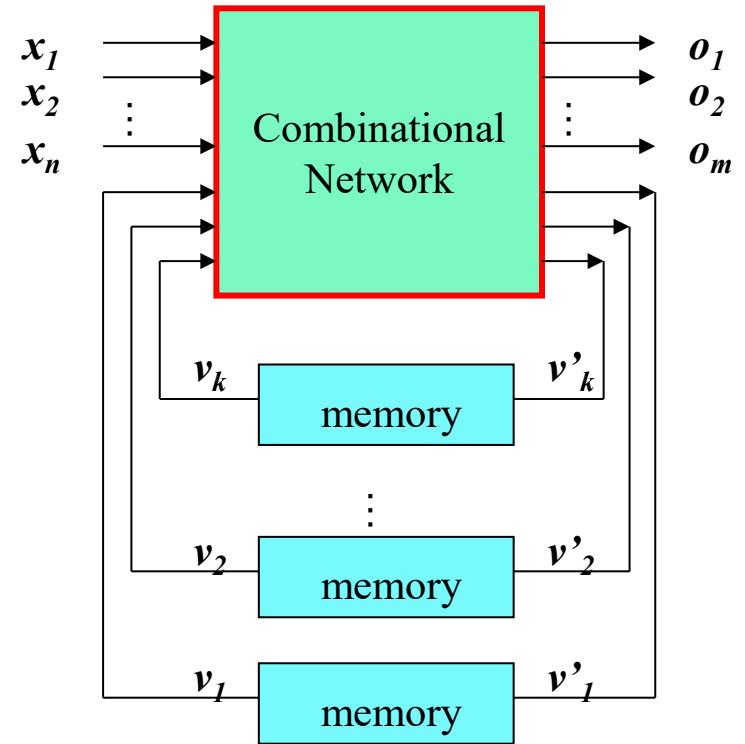


A Road Map

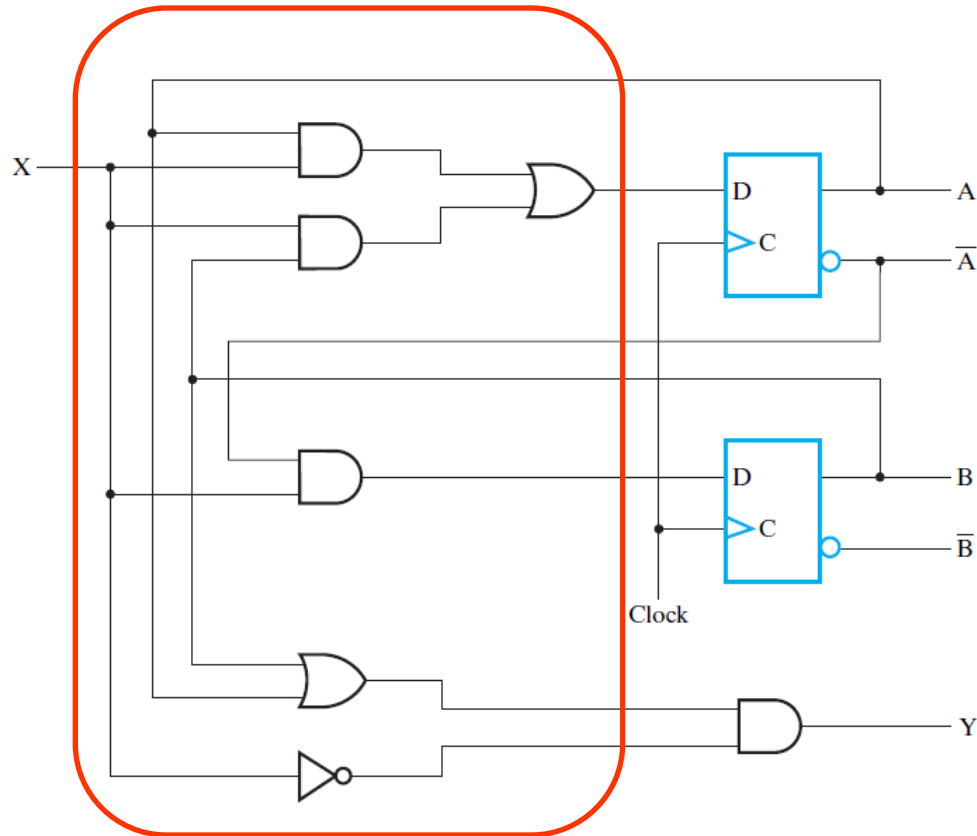


Symbolic Representation

- A sequential system has two parts
 - Combinational part
 - Sequential part



Representation of Combinational Part



Represent it?



c17 ISCAS Example

```
# c17 iscas example
# total number of lines in the netlist .....17
# lines from primary input gates ..... 5
# lines from primary output gates ..... 2
# lines from interior gate outputs ..... 4
# lines from ** 3 ** fanout stems ...6
# avg_fanin = 2.00, max_fanin = 2
# avg_fanout = 2.00, max_fanout = 2
```

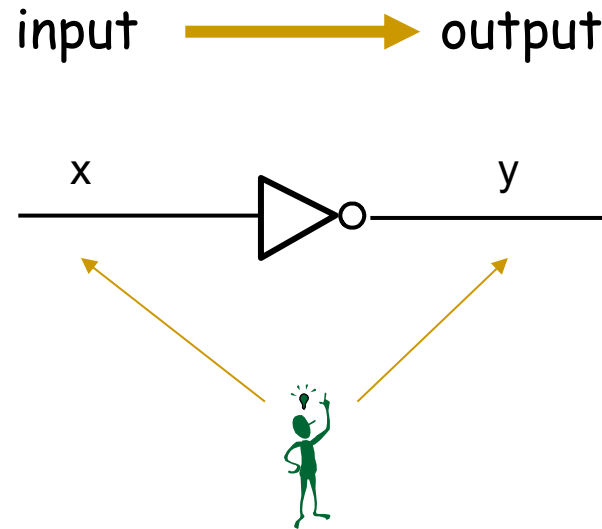
- INPUT(G1gat)
- INPUT(G2gat)
- INPUT(G3gat)
- INPUT(G6gat)
- INPUT(G7gat)
- OUTPUT(G22gat)
- OUTPUT(G23gat)

- G10gat = nand(G1gat, G3gat)
- G11gat = nand(G3gat, G6gat)
- G16gat = nand(G2gat, G11gat)
- G19gat = nand(G11gat, G7gat)
- G22gat = nand(G10gat, G16gat)
- G23gat = nand(G16gat, G19gat)

A Circuit Netlist: Combinational c432.bench of 250 Gates

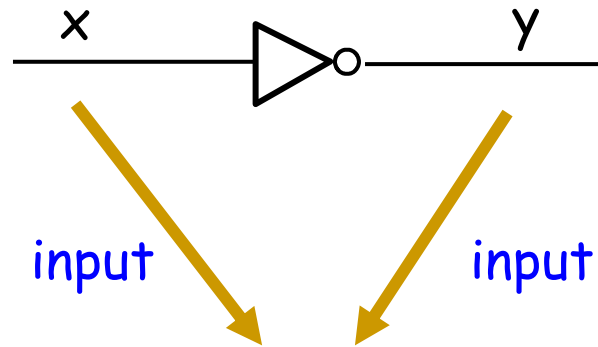
INPUT(G1)	G118 = NOT(G1)	G197 = NOR(G35,G151)	# G242 = XOR(G203,G174)	G294 = NAND(G250,G198)	Y333 = NAND(G272,W333)	G355 =
INPUT(G2)	G119 = NOT(G2)	G198 = NOR(G36,G151)	W242 = NOT(G203)	G295 =	G333 = NAND(X333,Y333)	AND(G346,G347,G348,G349,G350,G35
INPUT(G3)	G122 = NOT(G4)	G199 =	Z242 = NOT(G174)	AND(G259,G263,G266,G269,G	G334 = NAND(G318,G7)	1,G352,G353,G354)
INPUT(G4)	G123 = NOT(G6)	AND(G154,G159,G162,G165,G16	X242 = NAND(G203,Z242)	272,G275,G278,G281,G284)	# G335 = XOR(G308,G275)	G358 = NOT(G355)
INPUT(G5)	G126 = NOT(G8)	8,G171,G174,G177,G180)	Y242 = NAND(G174,W242)	G299 = NOT(G262)	W335 = NOT(G308)	G368 = NAND(G5,G358)
INPUT(G6)	G127 = NOT(G10)	G203 = NOT(G199)	G242 = NAND(X242,Y242)	G300 = NOT(G287)	Z335 = NOT(G275)	G369 = NAND(G358,G9)
INPUT(G7)	G130 = NOT(G12)	G213 = NOT(G199)	G245 = NAND(G213,G4)	G301 = NOT(G288)	X335 = NAND(G308,Z335)	G370 = NAND(G358,G13)
INPUT(G8)	G131 = NOT(G14)	# G223 = XOR(G203,G154)	# G246 = XOR(G203,G177)	G302 = NOT(G289)	Y335 = NAND(G275,W335)	G371 = NAND(G358,G17)
INPUT(G9)	G134 = NOT(G16)	W223 = NOT(G203)	W246 = NOT(G203)	G303 = NOT(G290)	G335 = NAND(X335,Y335)	G372 = NAND(G358,G21)
INPUT(G10)	G135 = NOT(G18)	Z223 = NOT(G154)	Z246 = NOT(G177)	G304 = NOT(G291)	G336 = NAND(G318,G11)	G373 = NAND(G358,G25)
INPUT(G11)	G138 = NOT(G20)	X223 = NAND(G203,Z223)	X246 = NAND(G203,Z246)	G305 = NOT(G292)	# G337 = XOR(G308,G278)	G374 = NAND(G358,G29)
INPUT(G12)	G139 = NOT(G22)	Y223 = NAND(G154,W223)	Y246 = NAND(G177,W246)	G306 = NOT(G293)	W337 = NOT(G308)	G375 = NAND(G358,G33)
INPUT(G13)	G142 = NOT(G24)	G223 = NAND(X223,Y223)	G246 = NAND(X246,Y246)	G307 = NOT(G294)	Z337 = NOT(G278)	G376 = NAND(G358,G36)
INPUT(G14)	G143 = NOT(G26)	# G226 = XOR(G203,G159)	G249 = NAND(G213,G8)	G308 = NOT(G295)	X337 = NAND(G308,Z337)	G377 = NAND(G2,G241,G332,G368)
INPUT(G15)	G146 = NOT(G28)	W226 = NOT(G203)	# G250 = XOR(G203,G180)	G318 = NOT(G295)	Y337 = NAND(G278,W337)	G378 = NAND(G245,G334,G369,G6)
INPUT(G16)	G147 = NOT(G30)	Z226 = NOT(G159)	W250 = NOT(G203)	# G328 = XOR(G308,G259)	G337 = NAND(X337,Y337)	G383 = NAND(G249,G336,G370,G10)
INPUT(G17)	G150 = NOT(G32)	X226 = NAND(G203,Z226)	Z250 = NOT(G180)	W328 = NOT(G308)	G338 = NAND(G318,G15)	G390 = NAND(G253,G338,G371,G14)
INPUT(G18)	G151 = NOT(G34)	Y226 = NAND(G159,W226)	X250 = NAND(G203,Z250)	Z328 = NOT(G259)	# G339 = XOR(G308,G281)	G396 = NAND(G254,G340,G372,G18)
INPUT(G19)	G154 = NAND(G118,G2)	G226 = NAND(X226,Y226)	Y250 = NAND(G180,W250)	X328 = NAND(G308,Z328)	W339 = NOT(G308)	G401 = NAND(G255,G342,G373,G22)
INPUT(G20)	G157 = NOR(G3,G119)	# G229 = XOR(G203,G162)	G250 = NAND(X250,Y250)	Y328 = NAND(G259,W328)	Z339 = NOT(G281)	G404 = NAND(G256,G343,G374,G26)
INPUT(G21)	G158 = NOR(G5,G119)	W229 = NOT(G203)	G253 = NAND(G213,G12)	G328 = NAND(X328,Y328)	X339 = NAND(G308,Z339)	G408 = NAND(G257,G344,G375,G30)
INPUT(G22)	G159 = NAND(G122,G6)	Z229 = NOT(G162)	G254 = NAND(G213,G16)	# G329 = XOR(G308,G263)	Y339 = NAND(G281,W339)	G411 = NAND(G258,G345,G376,G34)
INPUT(G23)	G162 = NAND(G126,G10)	X229 = NAND(G203,Z229)	G255 = NAND(G213,G20)	W329 = NOT(G308)	G339 = NAND(X339,Y339)	G412 = NOT(G377)
INPUT(G24)	G165 = NAND(G130,G14)	Y229 = NAND(G162,W229)	G256 = NAND(G213,G24)	Z329 = NOT(G263)	G340 = NAND(G318,G19)	G413 =
INPUT(G25)	G168 = NAND(G134,G18)	G229 = NAND(X229,Y229)	G257 = NAND(G213,G28)	X329 = NAND(G308,Z329)	# G341 = XOR(G308,G284)	AND(G378,G383,G390,G396,G401,G40
INPUT(G26)	G171 = NAND(G138,G22)	# G232 = XOR(G203,G165)	G258 = NAND(G213,G32)	Y329 = NAND(G263,W329)	W341 = NOT(G308)	4,G408,G411)
INPUT(G27)	G174 = NAND(G142,G26)	W232 = NOT(G203)	G259 = NAND(G223,G157)	G329 = NAND(X329,Y329)	Z341 = NOT(G284)	G414 = NOT(G390)
INPUT(G28)	G177 = NAND(G146,G30)	Z232 = NOT(G165)	G262 = NAND(G223,G158)	# G330 = XOR(G308,G266)	X341 = NAND(G308,Z341)	G415 = NOT(G401)
INPUT(G29)	G180 = NAND(G150,G34)	X232 = NAND(G203,Z232)	G263 = NAND(G226,G183)	W330 = NOT(G308)	Y341 = NAND(G284,W341)	G416 = NOT(G404)
INPUT(G30)	G183 = NOR(G7,G123)	Y232 = NAND(G165,W232)	G266 = NAND(G229,G185)	Z330 = NOT(G266)	G341 = NAND(X341,Y341)	G417 = NOT(G408)
INPUT(G31)	G184 = NOR(G9,G123)	G232 = NAND(X232,Y232)	G269 = NAND(G232,G187)	X330 = NAND(G308,Z330)	G342 = NAND(G318,G23)	G418 = NAND(G383,G414)
INPUT(G32)	G185 = NOR(G11,G127)	# G235 = XOR(G203,G168)	G272 = NAND(G235,G189)	Y330 = NAND(G266,W330)	G343 = NAND(G318,G27)	G421 = NAND(G383,G390,G415,G396)
INPUT(G33)	G186 = NOR(G13,G127)	W235 = NOT(G203)	G275 = NAND(G238,G191)	G330 = NAND(X330,Y330)	G344 = NAND(G318,G31)	G424 = NAND(G396,G390,G416)
INPUT(G34)	G187 = NOR(G15,G131)	Z235 = NOT(G168)	G278 = NAND(G242,G193)	# G331 = XOR(G308,G269)	G345 = NAND(G318,G35)	G425 = NAND(G383,G390,G404,G417)
INPUT(G35)	G188 = NOR(G17,G131)	X235 = NAND(G203,Z235)	G281 = NAND(G246,G195)	W331 = NOT(G308)	G346 = NAND(G328,G299)	G426 = NOT(G199)
INPUT(G36)	G189 = NOR(G19,G135)	Y235 = NAND(G168,W235)	G284 = NAND(G250,G197)	Z331 = NOT(G269)	G347 = NAND(G329,G300)	G427 = NOT(G295)
OUTPUT(G426)	G190 = NOR(G21,G135)	G235 = NAND(X235,Y235)	G287 = NAND(G226,G184)	X331 = NAND(G308,Z331)	G348 = NAND(G330,G301)	G428 = NOT(G355)
OUTPUT(G427)	G191 = NOR(G23,G139)	# G238 = XOR(G203,G171)	G288 = NAND(G229,G186)	Y331 = NAND(G269,W331)	G349 = NAND(G331,G302)	G429 = NOR(G412,G413)
OUTPUT(G428)	G192 = NOR(G25,G139)	W238 = NOT(G203)	G289 = NAND(G232,G188)	G331 = NAND(X331,Y331)	G350 = NAND(G333,G303)	G430 = NAND(G378,G383,G418,G396)
OUTPUT(G429)	G193 = NOR(G27,G143)	Z238 = NOT(G171)	G290 = NAND(G235,G190)	G332 = NAND(G3,G318)	G351 = NAND(G335,G304)	G431 = NAND(G378,G383,G421,G424)
OUTPUT(G430)	G194 = NOR(G29,G143)	X238 = NAND(G203,Z238)	G291 = NAND(G238,G192)	# G333 = XOR(G308,G272)	G352 = NAND(G337,G305)	G432 = NAND(G378,G418,G421,G425)
OUTPUT(G431)	G195 = NOR(G31,G147)	Y238 = NAND(G171,W238)	G292 = NAND(G242,G194)	W333 = NOT(G308)	G353 = NAND(G339,G306)	
OUTPUT(G432)	G196 = NOR(G33,G147)	G238 = NAND(X238,Y238)	G293 = NAND(G246,G196)	Z333 = NOT(G272)	G354 = NAND(G341,G307)	
		G241 = NAND(G1,G213)		X333 = NAND(G308,Z333)		

Simulation View: $y \Leftarrow f(x)$



- Directed: y is function of x .

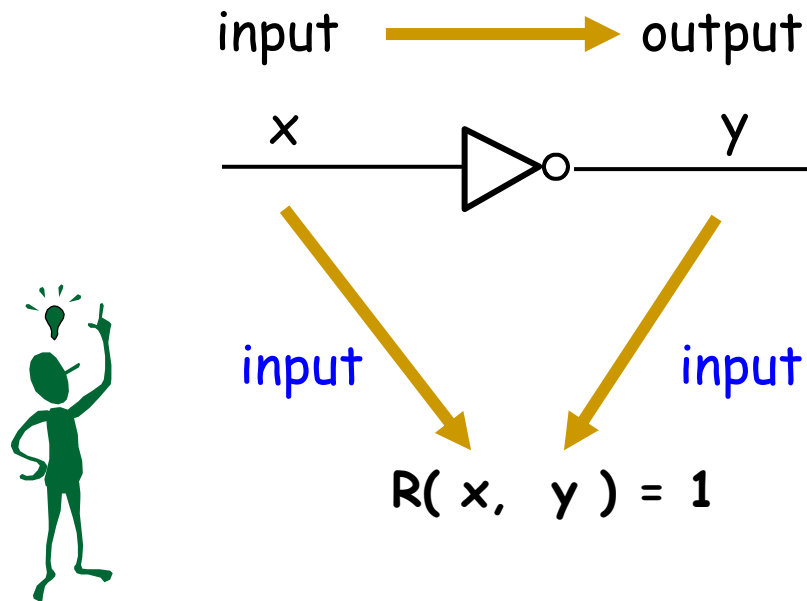
Third Party Observation



$$R(x, y) = 1$$

Characteristic Boolean Function

Method 1: Truth Table

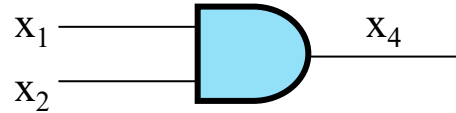


x	y
0	1
1	0



x	y	$R(x, y)$
0	1	1
1	0	1
0	0	0
1	1	0

Method 1: Truth Table



x_1	x_2	x_4
0	0	0
0	1	0
1	0	0
1	1	1



x_1	x_2	x_4	R
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	1
			0

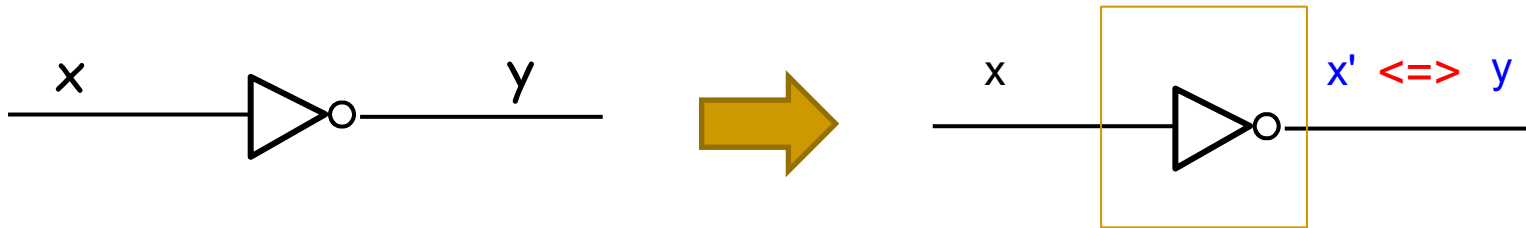
$$R_{\text{and1}} = \neg x_1 \neg x_2 \neg x_4 + \neg x_1 x_2 \neg x_4 + x_1 \neg x_2 \neg x_4 + x_1 x_2 x_4$$

Review: Boolean Function $F(P, Q) = (P \Leftrightarrow Q)$

P	Q	$P \Leftrightarrow Q$
0	0	1
0	1	0
1	0	0
1	1	1

- $P \Leftrightarrow Q = PQ + P'Q' = (P \oplus Q)' = P \otimes Q$
- When Boolean function $F(P, Q) = (P \Leftrightarrow Q)$ is 1, it implies $P=Q$

Method 2: Equivalence $\Leftrightarrow$

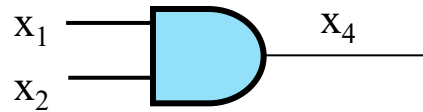


Equality =

Equivalence $\Leftrightarrow$

- From equality = to equivalence $\Leftrightarrow$
- Equality: $y = x'$
- Equivalence: $R(x, y) = (y \Leftrightarrow x')$
- If Characteristic Boolean Function $R(x, y)=1$
 - $\Rightarrow y \Leftrightarrow x'$
 - $\Rightarrow y = x'$

Characteristic Representation (CR)



x_1	x_2	x_4	R
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	1
			0

$$R_{\text{and1}} = \neg x_1 \neg x_2 \neg x_4 + \neg x_1 x_2 \neg x_4 + x_1 \neg x_2 \neg x_4 + x_1 x_2 x_4$$



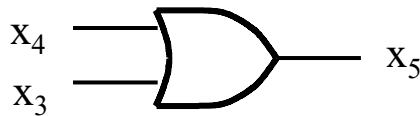
$$R_{\text{and1}} = R_{\text{and2}}$$

$$x_4 \Leftrightarrow (x_1 x_2)$$



$$R_{\text{and2}} = [x_4 \Leftrightarrow (x_1 x_2)]$$

Characteristic Representation (CR)



x_3	x_4	x_5	R
0	0	0	1
0	1	1	1
1	0	1	1
1	1	1	1
			0

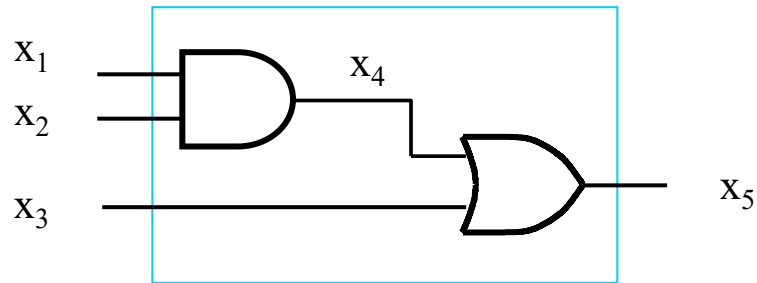
$$R_{or1} = \neg x_3 \neg x_4 \neg x_5 + \neg x_3 x_4 x_5 + x_3 \neg x_4 x_5 + x_3 x_4 x_5$$

$$R_{or1} = R_{or2}$$

$$x_5 \Leftrightarrow (x_3 + x_4)$$

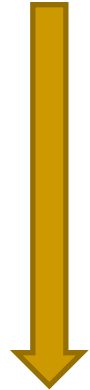
$$R_{or2} = [x_5 \Leftrightarrow (x_3 + x_4)]$$

CR with Truth Table



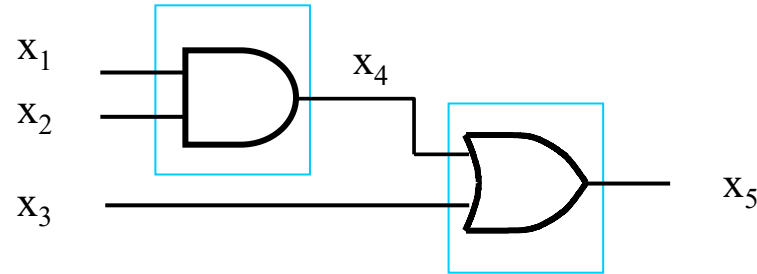
x_1	x_2	x_3	x_5
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1

x_1	x_2	x_3	x_5	R_1
0	0	0	0	1
0	0	1	1	1
0	1	0	0	1
0	1	1	1	1



■ $R_1 = ?$

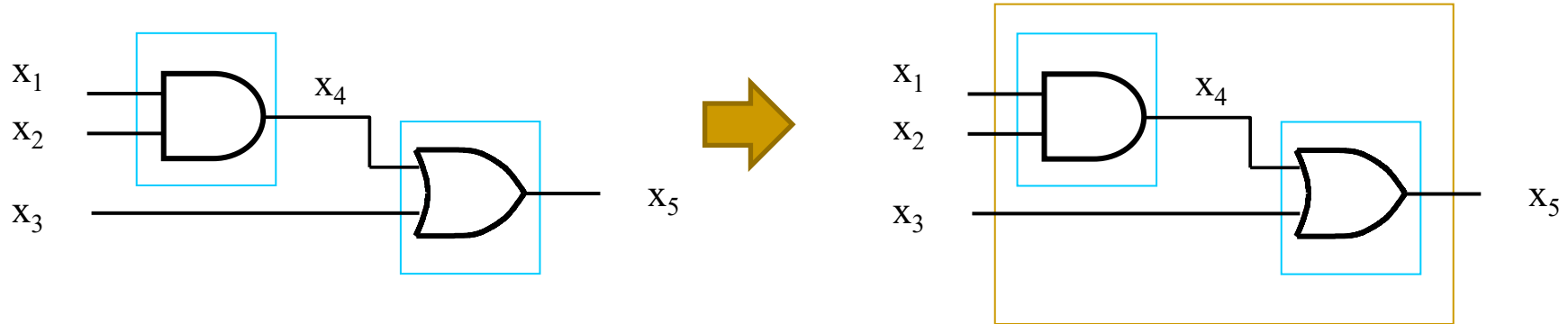
Compositional CR with Internal Variables



- $R_{\text{or}} = [x_5 \Leftrightarrow (x_3 + x_4)]$
- $R_{\text{and}} = [x_4 \Leftrightarrow (x_1 x_2)]$
- $R_2 = R_{\text{and}} \wedge R_{\text{or}}$
- $= [x_4 \Leftrightarrow (x_1 x_2)] \wedge [x_5 \Leftrightarrow (x_3 + x_4)]$

■ $R_1 = R_2$? If not, why?

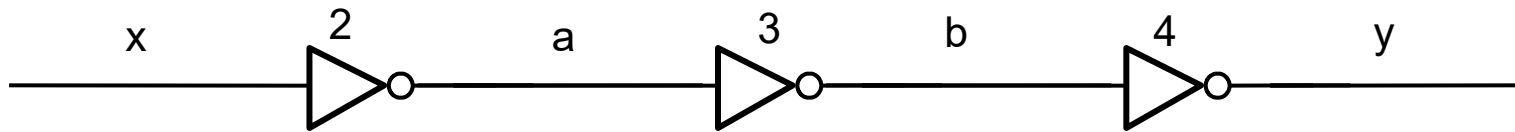
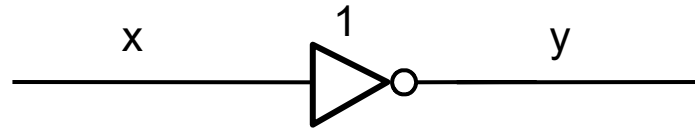
Compositional CR without Internal Variables



- $R_{\text{or}} = [x_5 \Leftrightarrow (x_3 + x_4)]$
- $R_{\text{and}} = [x_4 \Leftrightarrow (x_1 x_2)]$
- $R_3 = \exists x_4. (N_{\text{and}} \wedge N_{\text{or}})$
- $= \exists x_4. [x_4 \Leftrightarrow (x_1 x_2)] \wedge [x_5 \Leftrightarrow (x_3 + x_4)]$

- $R_1 = R_3 ?$

Compositional CR



- $R_1 = [y \Leftrightarrow x']$
- $R_2 = [a \Leftrightarrow x']$
- $R_3 = [b \Leftrightarrow a']$
- $R_4 = [y \Leftrightarrow b']$

- $R_1 = [y \Leftrightarrow x']$
- $R_2 = [a \Leftrightarrow x']$
- $R_3 = [b \Leftrightarrow a']$
- $R_4 = [y \Leftrightarrow b']$
- $R_k = R_2 \wedge R_3 \wedge R_4$
- $R_{234} = \exists a \exists b. (R_2 \wedge R_3 \wedge R_4)$

- $R_1 = R_k?$
- $R_1 = R_{234}?$

c17 ISCAS Example

- INPUT(G1gat)
- INPUT(G2gat)
- INPUT(G3gat)
- INPUT(G6gat)
- INPUT(G7gat)
- OUTPUT(G22gat)
- OUTPUT(G23gat)

- $G10gat = \text{nand}(G1gat, G3gat)$
- $G11gat = \text{nand}(G3gat, G6gat)$
- $G16gat = \text{nand}(G2gat, G11gat)$
- $G19gat = \text{nand}(G11gat, G7gat)$
- $G22gat = \text{nand}(G10gat, G16gat)$
- $G23gat = \text{nand}(G16gat, G19gat)$

- $G10gat = \text{nand}(G1gat, G3gat)$
- $C = \text{nand}(A, B)$
- $C \Leftrightarrow (AB)'$
- Characteristic Rep
 - $= [C \Leftrightarrow (AB)']$

References

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